Chapter summary

Familiarising yourself with **items in green** is important for effective revision.
Paying attention to items in the boxes is helpful for understanding the concept(s).

15.1 Definition of Gross Domestic Product (GDP)

- **Gross domestic product** (GDP) is **the① total market value of ②production of all ③resident producing units of ④a country or an economic territory in ⑤a given period (e.g. a month, a quarter or a year)**.
- **Resident producing units** refers to the production units that maintain their centre of economic interest in an economic territory, irrespective of their nationalities.
- Key points of the definition of GDP:
  - Total market value
  - Value of production (the production of final goods and services)
  - Production of a country or an economic territory
  - A given period

- The following values are not included in the calculation of GDP:
  - values that are not derived from current production
  - values of goods that are not sold to the market
  - values of second-hand goods
  - values of goods that are not produced by resident producing units

- Interests and dividends earned by lenders and shareholders are included in the calculation of GDP.

15.2 Calculation of GDP

Introduction to the three approaches of measuring GDP

- Illustration using the circular flow model:
  - The two basic economic units: households and firms
  - Real flow and money flow:

Real flow (circular flow of goods and services)	Money flow (circular flow of money)
Firms make use of the factors of production offered by the households to produce final goods and services, which are sold to households.	The factor income earned by the households is the firms' cost of production. The firms' revenue is also the households' consumption expenditure.

- GDP calculated by the three approaches are theoretically the same.

Production approach	=	Expenditure approach	=	Income approach
Total market value of production		Total expenditure		Total income

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Production approach (Value added approach)

- The **production approach** (**value added approach**) measures **the total value of production** of resident producing units in a given period of time.
- The GDP calculated using the production approach is equal to the sum of value added of all recent producing units in a given period of time.
- Meaning of value added

$$\text{Value added} = \text{Gross value of output} - \text{Value of intermediate consumption}$$

- GDP measured **by the production approach** is known as **GDP at factor cost**.

When imported materials are involved in the production chain, their values should be deducted to find the value added of the resident producing units.

Expenditure approach

- The **expenditure approach** measures **the total expenditure on final goods and services** produced by resident producing units in a given period of time.
- GDP measured **by the expenditure approach** is known as **GDP at market prices**.
- The GDP calculated using the expenditure approach can be calculated using the following formula:

$$\begin{aligned}\text{GDP} = {} & \text{C (Private consumption expenditure)} + \text{I (Gross investment)} + \\ & \text{G (Government consumption expenditure)} + \text{NX (Net exports)}\end{aligned}$$

- **Private consumption expenditure** (C) refers to consumption expenditures by households and private non-profit institutions on currently produced final goods and services.
- **Gross investment** (I) refers to the expenditure made mainly by firms on final goods and services, especially on capital goods.

$$\begin{aligned}\text{Gross investment} &= \text{Gross domestic fixed capital formation} + \text{Changes in inventories} \\ &= \text{Net domestic fixed capital formation} + \text{Depreciation} + \text{Changes in inventories}\end{aligned}$$

The value of unsold final goods produced in the current year is included in GDP as changes in inventories.

- **Government consumption expenditure** (G) includes government purchases of goods and services and payments made to government employees.
- **Net exports** (NX) refers to the difference between the value of total exports (X) and the value of total imports (M) of goods and services.

$$\text{Net exports (NX)} = \text{Total exports of goods and services (X)} - \text{Total imports of goods and services (M)}$$

Transfer payments provided by the government are not included in government consumption expenditure.

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Income approach

- The **income approach** measures GDP by summing the income received by factors of production from providing factor service. It is equal to the sum of compensation of employees and gross operating surplus of firms.



Components of the GDP compiled under the income approach are not required in the curriculum.

15.3 GDP at Market Prices and GDP at Factor Cost

- GDP at market prices may be different from GDP at factor cost. The difference can be due to indirect taxes and subsidies.
- The difference can be expressed using the following formula:

GDP at market prices = GDP at factor cost + Indirect taxes – Subsidies

- Indirect taxes and subsidies

Indirect taxes	Subsidies
With indirect taxes, the amount paid by consumers (i.e. GDP at market prices) is higher than the amount received by factor owners (i.e. GDP at factor cost)	With subsidies, the amount paid by consumers (i.e. GDP at market prices) is lower than the amount received by factor owners (i.e. GDP at factor cost)

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